

Separating the effects of strategy and hydroclimate in COEQWAL results

§2 WHAT THIS PAGE IS

Every COEQWAL scenario combines two components: a management strategy and a hydroclimate. The strategy is the set of water management choices people can make. The hydroclimate is the pattern of temperature, precipitation, and flow conditions that no one controls. When a modeled result differs from one scenario to another, either component, or both, can be behind the difference.

This page explains how that difference can be separated into a part associated with the hydroclimate and a part associated with the management strategy. It walks through simplified illustrations first, then shows one example from actual modeled results.

This page explains a method for reading modeled results from the COEQWAL ensemble; the numeric values in its illustrations are invented for teaching and the single worked example reports one modeled case. Splitting a difference into parts does not indicate whether any part, or the difference itself, is acceptable. COEQWAL does not recommend any course of action.

§3 TWO COMPONENTS, ONE BASELINE

All comparisons start from one scenario, the baseline: current operations under the historical hydroclimate. It is the reference point every other result is read against. Every other scenario in the COEQWAL library changes the strategy, the hydroclimate, or both, and a scenario's difference from the baseline splits into two parts. The split holds operations fixed first: it uses current operations under the scenario's hydroclimate as a second reference point, so that each strategy is compared with current operations under the same hydroclimate.

The hydroclimate effect is the difference between current operations under that hydroclimate and the baseline. It answers: what changes if operations stay the same but the hydroclimate is different? The management effect is the difference between the scenario and current operations under that same hydroclimate. It answers: what changes if the hydroclimate is the same but the management strategy is different? The two effects add exactly to the scenario's difference from the baseline: $\text{baseline value} + \text{hydroclimate effect} + \text{management effect} = \text{scenario value}$. The summary briefs print an effect smaller than 0.1 as 0.0, so a printed table row can differ from the sum by less than 0.1 while the continuous values add exactly.

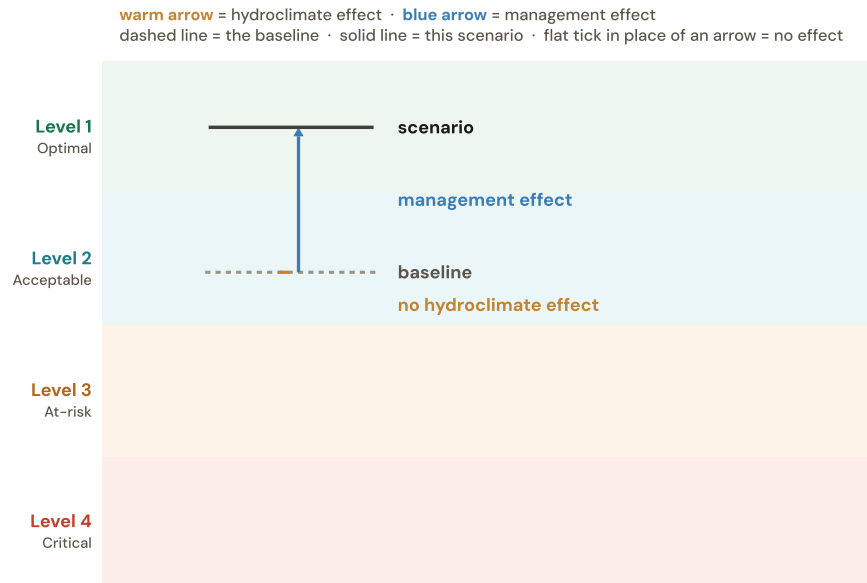
Two consequences follow. If a scenario keeps current operations, its management effect is zero; if it keeps the historical hydroclimate, its hydroclimate effect is zero. And because the management effect is measured under the scenario's own hydroclimate, the same strategy can have a different management effect under different hydroclimates. That variation is how the interplay between strategy and hydroclimate shows up in results, and it is why the briefs report the management effect separately for each hydroclimate.

Three notes for careful readers. First, the split applies only to continuous outcome values, not to outcome level categories. When a result crosses a level boundary, the crossing belongs to the whole difference from the baseline and is not

attributed to the management strategy alone or to the hydroclimate alone. Second, the split could be built the other way round, measuring the strategy first under the historical hydroclimate; COEQWAL uses the route through current operations so that every strategy is compared with current operations under the same hydroclimate. Third, the split describes where a modeled difference comes from. It does not say whether the difference matters, which is a judgment the framework leaves to the reader.

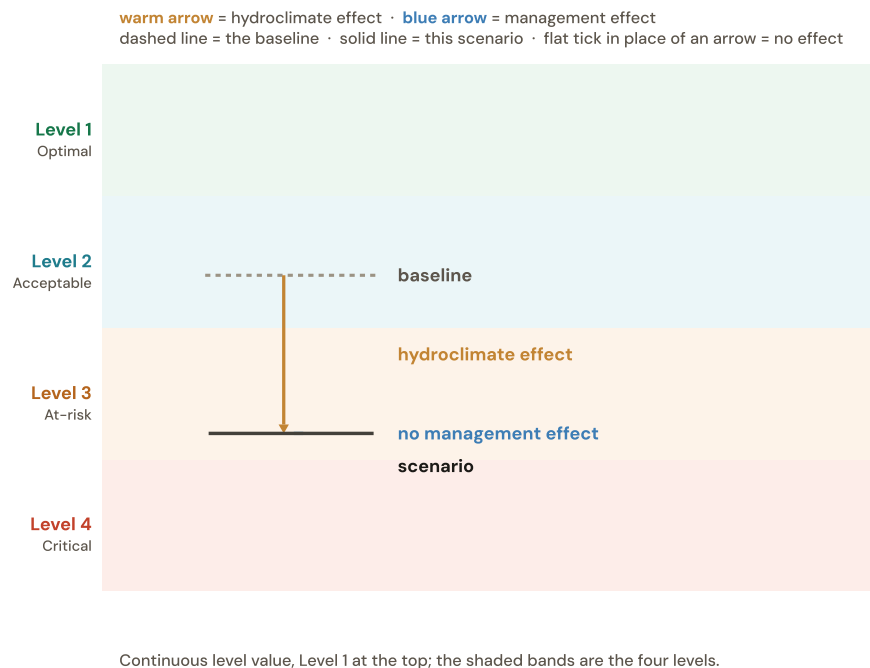
§4 WHAT THE SPLIT CAN LOOK LIKE

The values in the five panels below are invented for illustration. They are not modeled results. Each panel is drawn the way the summary briefs draw their figures: Level 1 at the top, the baseline as a dashed line, a warm arrow for the hydroclimate effect, a blue arrow for the management effect, and a solid mark where the scenario value lands.

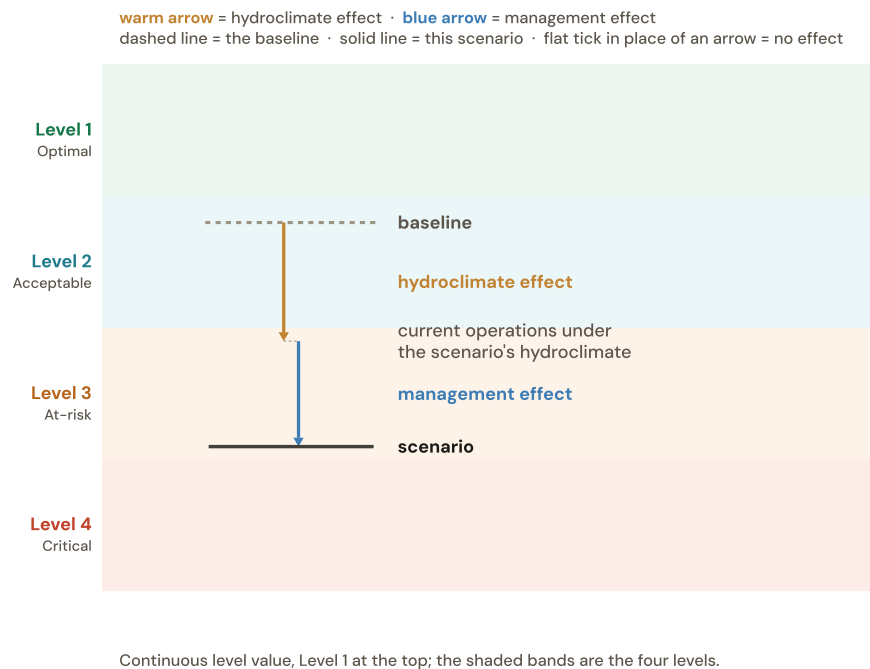


Continuous level value, Level 1 at the top; the shaded bands are the four levels.

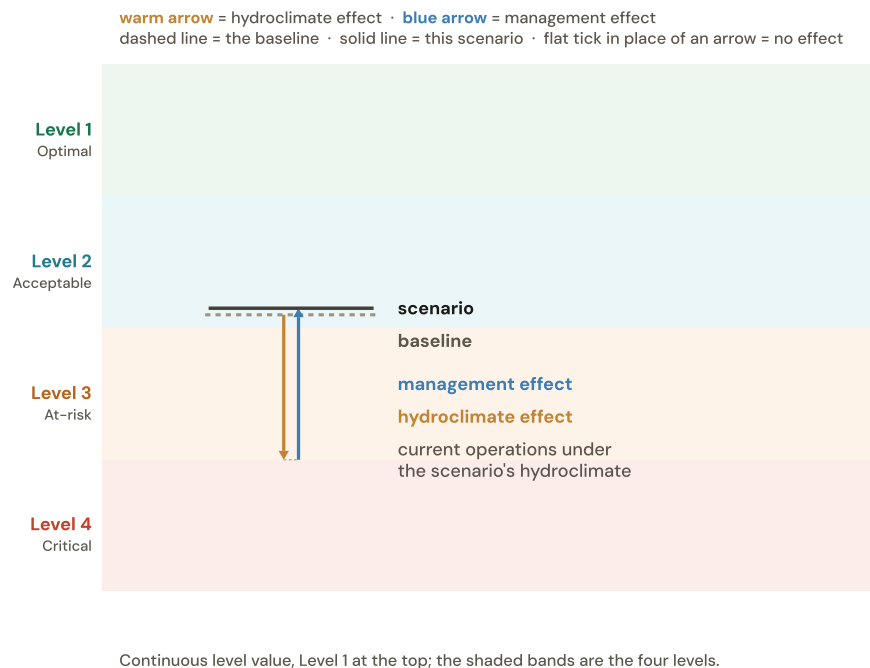
Panel 1, management effect only. The scenario keeps the historical hydroclimate, so there is no hydroclimate arrow. The management arrow carries the whole difference from the baseline.



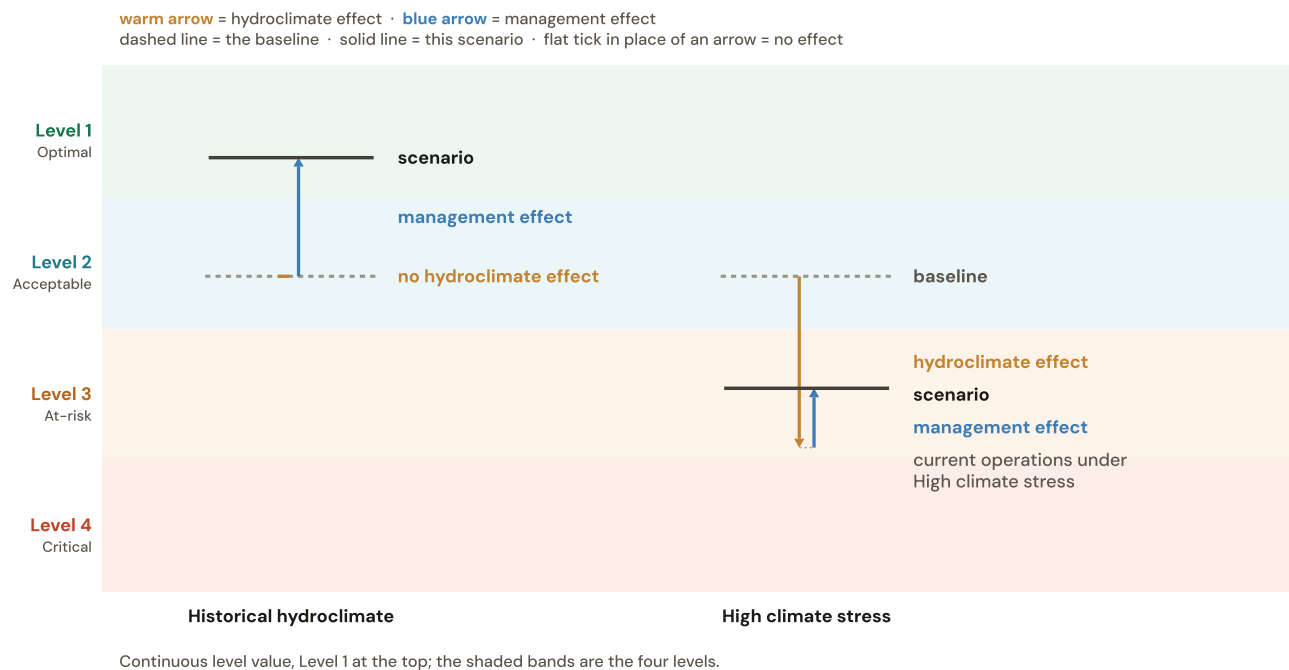
Panel 2, hydroclimate effect only. The scenario keeps current operations under a different hydroclimate. The hydroclimate arrow carries the whole difference and there is no management arrow.



Panel 3, both effects, same direction. The two arrows point the same way and the parts add. The scenario value differs from the baseline by the sum of the two effects.



Panel 4, both effects, opposite directions. The arrows point opposite ways and largely offset. The scenario value lands near the baseline even though both parts are sizeable. This is the case worth pausing on: a result that looks nearly unchanged can contain two substantial parts pulling against each other.

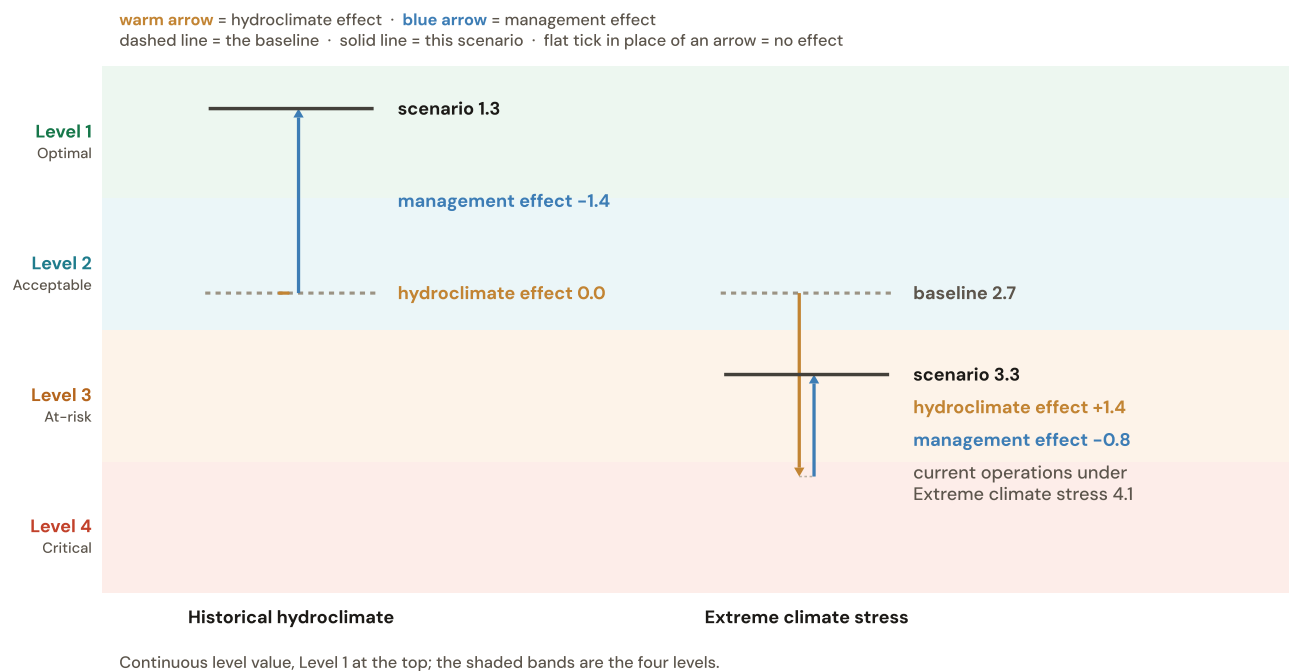


Panel 5, same strategy, two hydroclimates. Two lanes side by side, the same strategy in each. Under the historical hydroclimate there is only a management arrow. Under a stressed hydroclimate the hydroclimate arrow appears, and the management arrow is a different length than it was in the first lane. The management effect of a strategy depends on the hydroclimate it is run under.

The offsetting case in Panel 4 is why some COEQWAL summary tables outline cells where the hydroclimate and management effects point in opposite directions and each is at least 0.1 on the level scale. An outlined cell is a signal that the single number shown is the net of two parts, not the whole story.

§5 ONE EXAMPLE FROM ACTUAL RESULTS

The clean panels above are teaching cases. Actual modeled results are usually messier: both effects present, and the management effect changing size from one hydroclimate to the next. The figure below shows one actual result from the COEQWAL ensemble, drawn as two lanes.



In CalSim 3 simulations, freshwater for Delta exports at Banks Pumping Plant sits at 2.7 under the baseline. Under the historical hydroclimate, winter-run refuge flows shift the value by -1.4 (the management effect; there is no hydroclimate effect). Under Extreme climate stress, the hydroclimate shifts the value by +1.4 under current operations (the hydroclimate effect) and winter-run refuge flows shift it by a further -0.8 (the management effect), landing at 3.3. The management effect is -1.4 under one hydroclimate and -0.8 under the other. This example was chosen to show a case in which both effects are visible and the management effect differs between hydroclimates.

§6 WHERE THIS APPEARS IN COEQWAL

Readers will encounter this idea in the summary briefs on the Data page, where outlined table cells mark offsetting effects in the two-part form described above, and in comparisons across the Explore section, where the same result can be read against different strategies or different hydroclimates. The decomposition is a way of reading those materials, not a separate dataset.

§7 REFERENCES

The worked example in §5 is read from the COEQWAL data snapshot of August 29, 2026, at the continuous value the model reports rather than at the level band it falls in. Values are shown to one decimal, floored rather than rounded, and each effect is the difference between the one-decimal values, so the two effects and the baseline add exactly to the scenario value as printed. The four CalSim 3 runs behind the example, and the internal codes for the key outcome and the location, are listed here for citation. They are internal identifiers, not public names.

CELL OF THE EXAMPLE	RUN
Current operations, Historical hydroclimate	s0020
Winter-run refuge flows, Historical hydroclimate	s0031
Current operations, Extreme climate stress	s0108
Winter-run refuge flows, Extreme climate stress	s0118
Key outcome, Freshwater for Delta exports	FW_EXP
Location, Banks Pumping Plant	CAA003

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Supported by the California Climate Action Initiative • UC Grant R02CM7222

Updated September 11, 2026

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